IILM UNIVERSITY, GREATER NOIDA | SUMMER INTERNSHIP PRESENTATION 2026

PYTHON DEVELOPER INTERNSHIP

URL Shortener Application (GUI + Persistent Storage)

Student Name: Prince Maurya

Roll No.: 25SCS1003001538 | **Section:** 2CSE7 (3rd Sem)

Programme: B.Tech Computer Science & Engineering (2025-2029)

Industry Partners: upSkill Campus | UniConverge Technologies (UCT) | AICTE

Certificate ID: USC838112TIA (12-08-2026 to 02-09-2026)

01. ABOUT THE STUDENT

ACADEMIC & INTERNSHIP PROFILE

Academic Details

Student Name: Prince Maurya

Roll Number: 25SCS1003001538

Department: CSE, IILM University

Batch: 2025-2029 (3rd Semester)

Section: 2CSE7

Internship Credentials

Domain: Python Programming

Role: Python Developer Intern

Partners: upSkill Campus + UCT + AICTE

Duration: 12-08-2026 to 02-09-2026

Cert ID: `USC838112TIA`

02. INTERNSHIP OBJECTIVES

LEARNING GOALS & DELIVERABLES

- **Core Python Mastery:** Strengthen fundamentals, logic building, data structures, and modular function definitions.
- **Event-Driven Programming:** Build an interactive Desktop GUI using Python's `tkinter` library.
- **Data Persistence:** Master local file I/O operations and JSON serialization using the `json` module for permanent storage.
- **Defensive Programming:** Prevent crashes using rigorous `try-except` blocks and input validation checks.

03. LEARNING PROGRESS & MILESTONES

4-WEEK STRUCTURED ROADMAP

SPRINT	FOCUS AREA	KEY DELIVERABLES
Week 01	**Python Core & Logic**	Selected project; developed initial CLI code using `string` & `random` libraries.
Week 02	**Data Persistence**	Integrated `json` file storage (`url_database.json`) and added HTTP/HTTPS validation.
Week 03	**GUI Development**	Designed desktop window interface using `tkinter`; integrated OS clipboard copy functionality.
Week 04	**Testing & Submission**	Added `messagebox` error alerts, "Clear All" button, completed end-to-end testing, pushed to GitHub.

04. PROJECT OVERVIEW: URL SHORTENER

PROBLEM & SOLUTION ARCHITECTURE

The Problem

Long, cumbersome web URLs are hard to share in documents, presentation slides, or emails. Public cloud solutions (like Bitly) require paid tiers, track users, and require constant internet connectivity to manage links.

The Desktop Solution

A Python desktop application with a clean GUI that converts long links into unique 6-character short codes, backed by persistent local JSON storage ensuring privacy, speed, and 100% offline usability.

05. TECHNOLOGY STACK

CORE TOOLS & MODULES

💻 **Python 3.x:** Main language & backend logic

🔒 **String & Random:** 6-char code generation

📞 **Tkinter:** GUI windows, inputs, buttons

📁 **OS Module:** File existence verification

06. SYSTEM ARCHITECTURE

END-TO-END EXECUTION FLOW



- **Validation Layer:** Ensures URL string starts with exactly `http://` or `https://`.
- **Algorithmic Layer:** Creates randomized 6-character alphanumeric combinations (`while` loop blocks collisions).
- **Database Layer:** Synchronizes Python dictionaries directly with `url_database.json`.

07. CORE APPLICATION FEATURES

FUNCTIONAL CAPABILITIES

- **Instant URL Shortening:** Generates unique 6-character links instantly.
- **Collision Prevention:** Automated loop prevents duplication of existing short codes.
- **Original URL Retrieval:** Enter short code to instantly fetch original long URL.
- **Clipboard Integration:** Single-click copy to system clipboard (`root.clipboard_append`).
- **Defensive Validation:** Pop-up alerts (`messagebox`) prevent empty or malformed inputs.

08. TESTING & VERIFICATION

QUALITY ASSURANCE CHECKS

TEST SCENARIO	OBSERVED BEHAVIOR	RESULT
Empty Input Submitted	Warning messagebox popup shown; application does not crash	PASS
Invalid URL Format	Format warning popup shown (Missing HTTP)	PASS
Clipboard Copy Clicked	Short link copied to OS memory successfully; ready to paste	PASS
Application Reboot	All previously shortened URLs persist in `url_database.json`	PASS

09. GITHUB SUBMISSION

REPOSITORY DETAILS

GitHub Repository Link

https://github.com/Princemau25/url-shortener

College Standard Naming Convention

Format: Session_Name_RollNo_Semester_Section

Submitted As: `2025-2029_Prince_Maurya_25SCS1003001538_3rd_sem_2CSE7`

10. CONCLUSION & FUTURE SCOPE

SUMMARY & ROADMAP

- **Key Takeaways:** Bridged Python fundamentals with real-world software design. Mastered Tkinter GUI building, JSON data persistence, defensive programming, and professional GitHub documentation.
- **Future Enhancements:**
  - Migrate local JSON storage to cloud databases (Firebase / MongoDB).
  - Refactor GUI into a Flask / FastAPI web application.
  - Add link analytics (click counts & geographical metrics).